

Tide retrospective: *factored κ_3 identity for affine observables (potential-agnostic)*

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Abstract

For any 1D Gibbs measure with potential L at temperature t , with $Z(t) \neq 0$ and integrability of $x^n e^{-tL}$ for $n \in \{0, 1, 2, 3\}$, the joint third cumulant of three shifted-affine observables factors as

$$\kappa_3(\mu_{\text{vol}}, L, a_2x + b_2, t, a_1x + b_1, a_3x + b_3) = a_1 a_2 a_3 \cdot \kappa_3(\mu_{\text{vol}}, L, x, t, x, x).$$

This is the natural cumulant content GPT-5.5 Pro named at C1's deliberation: *constants drop by translation invariance, scalings factor by trilinearity*. The C1 (harmonic-affine vanishing) and G2/G2+ (anharmonic-affine asymptotic) tides become 5-line specialisations of this identity. New file `Laplace/AffineKappa3.lean`, 310 lines, four public theorems plus one private cubic-polynomial helper; zero `sorry`, zero `axiom`, zero `native_decide`; *single-attempt clean build with no mid-proof thrashing-rescue consult*.

1 Setting

The afternoon arc of 2026-05-07 has been dominated by the *I2 algebra programme* — one infrastructure tide (I2, generic `gibbsCov` algebra) followed by a stack of cheap strict-improvements off it (I3 affine-bilinear cov, C1 harmonic-affine κ_3 vanishing, G2 anharmonic-affine κ_3 asymptotic, G2+ shifted-affine extension). Each retrospective in the stack named the *factored κ_3 identity* as the natural cumulant-content abstraction: GPT identified it during C1's Step 2 deliberation; both C1's Follow-ups and G2's "Candidate C deferred" cited it; the v4 survey ranked it the top recommendation as the trilogy's narrative-closer.

This Tide closes that thread. The candidate is precisely the potential-agnostic version of the structural identity that C1 and G2/G2+ formalise as potential-specific corollaries. Once it lands, C1 and G2/G2+ can be refactored as 5-line specialisations (combining this identity with the potential-specific value of $\kappa_3(x, x, x)$ on the right-hand side); the refactor is a separate follow-up tide once C1 and G2/G2+ both reach `main`.

The Step 2 deliberation between Claude and GPT-5.5 Pro converged in a single round on a refined "B-lite" plan: lift two private bridge lemmas from C1 and G2 to public, add a specialised `gibbsExpectation_affine` helper, then derive the headline by direct unfold + cubic expansion. Both models voted the same plan; GPT additionally trimmed the public API away from generic translation-invariance (whose hypothesis bundle "gets annoyingly large") toward an affine-specialised approach.

The markdown tide `log`¹ carries the full deliberation; the GPT consult is preserved at the sibling `gpt_responses_q1/q1_v1.md`.

¹In the SRI repo at `projects/primer/tide-log/`, file `2026-05-07-tide-factored-kappa3-affine.md`.

2 The thing formalised

For any 1D Gibbs measure with potential $L : \mathbb{R} \rightarrow \mathbb{R}$ at temperature $t > 0$, given $Z(t) := \int e^{-tL} \neq 0$ and integrability of $x^n e^{-tL}$ for $n \in \{0, 1, 2, 3\}$, and any six constants $a_1, a_2, a_3, b_1, b_2, b_3 \in \mathbb{R}$,

$$\kappa_3(\mu_{\text{vol}}, L, a_2x + b_2, t, a_1x + b_1, a_3x + b_3) = a_1a_2a_3 \cdot \kappa_3(\mu_{\text{vol}}, L, x, t, x, x).$$

The Lean signature²:

```
theorem kappa3_volume_shifted_affine_eq_smul
  (L : ℝ → ℝ) {t : ℝ}
  (hZ : partitionFunction L t ≠ 0)
  (h_int_1 : Integrable (fun x => Real.exp (-(t * L x))))
  (h_int_x : Integrable (fun x => x * Real.exp (-(t * L x))))
  (h_int_x2 : Integrable (fun x => x^2 * Real.exp (-(t * L x))))
  (h_int_x3 : Integrable (fun x => x^3 * Real.exp (-(t * L x))))
  (a a a b b b : ℝ) :
  Threepoint.kappa3 (volume : Measure ℝ) L
    (fun x => a * x + b) t
    (fun x => a * x + b) (fun x => a * x + b)
  = a * a * a * Threepoint.kappa3 (volume : Measure ℝ) L
    (fun x => x) t (fun x => x) (fun x => x)
```

The argument-ordering convention follows G2/G2+ (a_2 in the A-slot, the perturbation direction of `Threepoint.kappa3`'s 6-argument signature). GPT explicitly recommended preserving this for consistency with the existing C1/G2/G2+ theorems; mathematically κ_3 is symmetric, but proving permutation lemmas at the formal level would be unnecessary cost.

The hypothesis bundle is the minimal set: $Z \neq 0$ for the constants to evaluate as themselves (`gibbsExpectation_const` returns 0 when $Z = 0$), and the four integrability conditions for the three `gibbsExpectation_add` splits in the cubic-polynomial helper. GPT confirmed at deliberation that no additional Gibbs-regularity hypothesis is needed beyond this; `Integrable` carries the measurability.

Three additional public theorems are exported alongside the headline:

- `threepoint_gibbsExp_volume_zero_eq` — the bridge from `Threepoint.gibbsExp` (perturbed Gibbs expectation, at $h = 0$, on volume) to `Laplace.gibbsExpectation`. Identical statement to the private version in C1's and G2's files; this lifts it to public.
- `kappa3_volume_unfold` — `Threepoint.kappa3` at volume unfolded into the standard five-term cumulant formula expressed in `Laplace.gibbsExpectation`. Lifted from G2's private; G2's version was already potential-agnostic.
- `gibbsExpectation_affine` — $\langle ax + b \rangle_t = a \langle x \rangle_t + b$ given $Z \neq 0$ plus integrability. GPT named this as the recommended affine-specialised helper over a generic translation-invariance lemma.

²`Laplace/AffineKappa3.lean, kappa3_volume_shifted_affine_eq_smul.`

3 Proof strategy

The proof is direct: unfold both sides of the headline via `kappa3_volume_unfold`, then for each of the seven `gibbsExpectation` terms appearing in the unfold (the cubic φAB product, three quadratic pairwise products, and three linear single-affines), expand the integrand into a cubic polynomial in x via `funext + ring` and reduce via the I2 algebra (`gibbsExpectation_smul`, `_add`, `_const`) to a linear combination of the four moments $m_0 = 1, m_1 = \langle x \rangle_t, m_2 = \langle x^2 \rangle_t, m_3 = \langle x^3 \rangle_t$.

Cubic-polynomial reduction. The cubic expansion is factored into a private helper for the moment-linear identity³:

$$\langle a_3x^3 + a_2x^2 + a_1x + a_0 \rangle_t = a_3m_3 + a_2m_2 + a_1m_1 + a_0,$$

under the hypothesis bundle.

Polynomial expansion of the seven integrands. For each of the seven $\langle \cdot \rangle$ terms in the LHS unfold, an inline `have` block establishes the moment-linear form. For example, the cubic product expands as

$$\begin{aligned} (a_1x + b_1)(a_2x + b_2)(a_3x + b_3) \\ = (a_1a_2a_3)x^3 + (a_1a_2b_3 + a_1b_2a_3 + b_1a_2a_3)x^2 \\ + (a_1b_2b_3 + b_1a_2b_3 + b_1b_2a_3)x + b_1b_2b_3. \end{aligned}$$

`funext + ring` closes the pointwise equality, then the cubic helper replaces the expectation with the moment-linear form. The three quadratic products are packaged as cubics with leading coefficient 0 to reuse the same helper.

Final collection. After all seven integrand expansions, the LHS is a polynomial in the three a_i , three b_i , and three moments m_1, m_2, m_3 . The RHS unfold gives $a_1a_2a_3(m_3 - 3m_2m_1 + 2m_1^3)$. `ring` closes the final equality.

4 Roadblocks and resolutions

There weren't any, in the sense that the proof landed in a single attempt: the very first commit had zero errors, zero warnings, and zero `sorry/axiom/native_decide`. This is the third tide today (after I4 and N1) where the Step 2 advisory carried sufficient tactical detail to make Step 3 mechanical. The pattern is consolidating into a recipe.

The closest thing to a roadblock was the architectural decision at Step 2 between three candidates of materially different shape:

- Candidate A: direct cubic-polynomial expansion (mirrors G2/G2+'s structure verbatim, ~200 lines).
- Candidate B: bilinearity plus translation-invariance lemmas (cleaner cumulant infrastructure but requires generic-observable hypotheses, ~150 lines).

³The content is just linearity of $\langle \cdot \rangle_t$ over polynomial integrands, but each `gibbsExpectation.add` call needs integrability of both summands, so the expansion is a sequence of three nested splits with cumulative-tail integrability witnesses built up from the four input monomial integrabilities.

- Candidate C: pure multilinearity over arbitrary observables (very clean but doesn’t subsume Q1, ~30 lines).

GPT’s verdict was *B-lite*: reuse I2’s expectation algebra (already in `Laplace.Gibbs`), add the specialised `gibbsExpectation_affine`, then derive the headline by direct unfold + cubic expansion (so in practice the proof body mirrors A’s structure, but uses I2’s algebra instead of a custom “polynomial-le-3 workhorse”).

GPT-5.5 Pro: I would not do full Candidate A unless you want a quick one-off proof with minimal API design. I would not stop at C, because it does not close Q1. Best target: B-lite — promote to public `threepoint.gibbsExp_volume_zero_eq` and `kappa3.volume_unfold`, prove a small reusable expectation API including `gibbsExpectation_affine`, then prove the specialized affine-id factorization theorem by one unfold + rewrites + `ring`.

The actual proof structure ended up closer to A than B-lite literally suggests — the headline does *not* use slot-by-slot multilinearity / translation invariance lemmas; it uses direct cubic expansion. This is because the integrand products $(a_1x + b_1)(a_2x + b_2)(a_3x + b_3)$ etc. are not affine-of-id, so an affine-specialised helper applies only to the three single-affine slot terms; the four product terms still need cubic-poly reduction. The `gibbsExp_cubic_eq` private helper handles all seven uniformly.

One small detail. GPT’s recommendation referenced `gibbsExpectation_add_const` as a possible alternative to `gibbsExpectation_affine`. We chose the affine form because the seven integrand terms in the unfold all reduce to “polynomial ≤ 3 in x ” form, and the affine helper covers exactly the three linear cases; the four cubic cases use the cubic helper directly. Adding a separate `_add_const` would not have shortened the proof.

Numerical sanity check pre-Step 3. Verified the identity at six anharmonic test cases (linear and shifted-affine, varying t and constants) against `scipy.integrate.quad`. All cases agreed to $\sim 10^{-13}$ relative error (numerical-integration precision); the identity holds *exactly*, not just asymptotically.⁴

5 What was Mathlib, what was new

Mathlib provided. `MeasureTheory.Integrable.const_mul` (for scalar multiplication of integrable functions), `MeasureTheory.Integrable.add` (for additive splits), and nothing else from Mathlib proper — the formalisation is entirely above the integrable-function/integral-linearity stack.

What was new. The Threepoint package and the I2 tide already provided the algebra; this tide adds the synthesis:

- `threepoint.gibbsExp_volume_zero_eq` — public promotion of a lemma that existed identically as private in C1’s and G2’s files. The promotion is the only structural delta; the proof itself is one `simp`.

⁴The script is preserved at `projects/primer/tide-log/gpt_responses_q1/q1_numerics.py`. The case with $a_1 = a_2 = a_3 = 1$ and large constants b_i is the *translation-invariance smoke test*: it confirms that the identity holds when only the constants vary, hence that the constants-drop part of the cumulant content is correct.

- `kappa3.volume.unfold` — public promotion of G2’s private. Unfolds the `Threepoint.kappa3` definition and applies the bridge seven times. Already potential-agnostic in G2’s version; this just exposes it.
- `gibbsExpectation.affine` — new lemma. Combines I2’s existing `_add`, `_smul`, and `_const`.
- `gibbsExp_cubic.eq` (private) — new cubic-polynomial workhorse. Three nested `_add` splits with cumulative-tail integrability bookkeeping. The potential-agnostic counterpart of C1’s harmonic-specific polynomial workhorse (which evaluated the moments to harmonic-specific values; here we leave them as $\langle x^k \rangle_t$).
- the headline factorisation theorem. The proof body is seven inline integrand-expansion blocks plus a final `ring`.

6 Lessons

- *The cumulative-tail integrability pattern.* The `gibbsExp_cubic.eq` helper needs ~ 6 `Integrable` witnesses (not just 4): the four input monomial integrabilities, plus integrabilities of the cumulative partial-sum tails for each `gibbsExpectation.add` step. Each tail is built one line via `funext + ring + .add` from the previous; the bookkeeping is mechanical but unavoidable. Worth recording as a template for future polynomial-Gibbs-expectation helpers.
- *The “package the integrand as a cubic with leading coefficient 0” trick.* The four quadratic and three linear integrands in the seven-term unfold are *not* naturally cubic. Casting them as cubics with leading coefficient 0 — i.e. expressing $a_1a_2x^2 + (a_1b_2 + a_2b_1)x + b_1b_2$ as $0 \cdot x^3 + (a_1a_2)x^2 + \dots$ — lets a single cubic helper handle all seven cases uniformly. The pattern is reused four times in this proof.
- *The Step 2 advisory continues to do all the work.* For the third consecutive tide today (I4, N1, Q1), the Step 2 GPT consult contained sufficient tactical detail that the proof landed single-attempt with no thrashing-rescue consult needed. The pattern is now: at Step 2, ask GPT specifically about hypothesis-bundle minimality, idiom choice (`ring-vs-mul_pow`, or here `gibbsExpectation.affine-vs-add_const`), and the expected proof-skeleton structure. When the consult addresses all three, Step 3 is mechanical.
- *The “centered-product identity” is a future tide candidate.* GPT named it as the shortest conceptual proof of Q1 (~ 45 lines) but recommended B-lite for long-term reusability. We took B-lite. The centered-product identity remains as a follow-up: it would be a clean alternative formulation of κ_3 ($\kappa_3(X, Y, Z) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)(Z - \mathbb{E}Z)]$) that could land separately and provide a second, more elegant proof of the affine identity.

7 Follow-ups

- *Refactor C1 and G2/G2+ to be 5-line specialisations of this identity.* Once C1 and G2/G2+ both reach `main` (C1 is already merged at 521acac; G2/G2+ awaits a merge), each existing potential-specific affine- κ_3 theorem becomes $\kappa_3(\dots) = a_1a_2a_3 \cdot \kappa_3(x, x, x)$ via this identity, then specialised to the potential-specific value of $\kappa_3(x, x, x)$ on the right-hand side. The harmonic case gives 0 (Tide 5’s vanishing), the anharmonic case gives the $-\alpha/\lambda^3$ asymptotic from Tide 9. Each refactor ~ 30 lines.

- *Centered-product identity for κ_3 .* GPT’s Candidate D, deferred from this tide. Statement: $\kappa_3(\mu, L, A, t, \varphi, B) = \langle (\varphi - \langle \varphi \rangle)(A - \langle A \rangle)(B - \langle B \rangle) \rangle_t$. About ~ 45 lines. Would provide a second elegant proof of the affine identity (the centered observables $(a_i x + b_i) - \langle a_i x + b_i \rangle = a_i(x - \langle x \rangle)$ pull through trilinearly).
- *Generic multilinearity-of- κ_3 over arbitrary observables.* G2’s ”Candidate C” deferred. Statement: $\kappa_3(\mu, L, c \cdot A_0, t, c \cdot \varphi_0, c \cdot B_0) = c^3 \cdot \kappa_3(\mu, L, A_0, t, \varphi_0, B_0)$. About ~ 30 lines. Strict-cleaner than the affine specialisation but doesn’t subsume Q1; a separate building block.
- *Promote `gibbsExp_cubic_eq` (and the affine version) to `lean-common`.* Once the C1/G2 refactors demonstrate the third and fourth consumer sites (the same helpers could land in the bounded-prior-monomial generalisation tide, candidate B2 in the survey), the cubic-polynomial workhorse is a natural `lean-common` target.
- *Threepoint-side analogue of I2’s `gibbsCov` algebra.* I3’s retrospective flagged this. The threepoint `gibbsCov` (perturbed-Gaussian, with $h \cdot A$ in the potential) has no algebra of its own yet; an analogous port would unblock h -dependent κ_3 tides off the Tide 13 thread.
- *κ_4 analogue.* Once a `kappa4` definition lands in Threepoint, the same argument should give κ_4 of four shifted-affines factoring as $a_1 a_2 a_3 a_4 \cdot \kappa_4(x, x, x, x)$. About ~ 120 lines: a degree-4 polynomial helper plus the eight-term unfold.
- *`\leanref`-tag the threepoint paper for this identity.* Once the next threepoint-paper tagging pass runs (K2 in the survey), tag the multilinearity-of-cumulants remark with `\leanref` pinned to this tide’s commit.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2: the verdict that B-lite (with the affine-specialised helper) is preferable to either full Candidate A’s degree-3 polynomial machinery or full Candidate B’s generic translation-invariance theorem; the unconditional vs. hZ-conditional distinction (scaling unconditional, translation-invariance needs $Z \neq 0$); and the cleaner Candidate D centered-product alternative deferred as follow-up. The full consult is preserved at `projects/primer/tide-log/gpt_responses_q1/q1_v1.md`. The proof landed single-attempt with no mid-proof thrashing-rescue consult needed.